Lecture 31 : PERT, Project Crashing

Introduction

In the dynamic world of project management, delivering projects on time and within budget is a constant challenge. Project managers often face scenarios where initial estimates prove insufficient, or external pressures demand accelerated timelines. This lecture delves into advanced techniques for managing project schedules, particularly when task durations are uncertain, and explores crucial aspects of team management that underpin project success. Understanding how to statistically estimate task times and strategically reduce project durations, known as "crashing," empowers managers to respond effectively to unforeseen circumstances and stakeholder demands.

Beyond the technicalities of scheduling, the human element remains paramount. A project's success is inextricably linked to the performance and cohesion of its team. This lecture also sheds light on foundational theories of organizational behavior and team dynamics, offering insights into motivating team members, resolving conflicts, and fostering a productive work environment. By integrating robust scheduling methodologies with effective team leadership, project managers can navigate complex projects with greater confidence and achieve optimal outcomes.

Core Concepts

Term	Definition	Significance
`PERT`	Program Evaluation and Review Technique, a project management method used when task durations are uncertain and statistical estimates are required.	Allows for probabilistic variations in activity durations, providing a more realistic project timeline in development work.
`Optimistic Time (a)`	The shortest possible time an activity can be completed, assuming everything goes perfectly without obstacles.	Represents the best-case scenario for task completion, used in PERT calculations.
`Most Likely Time (m)`	The most frequent or probable time an activity will take, under normal conditions.	Reflects the typical duration expected for a task, serving as a central estimate in PERT.

Term	Definition	Significance
`Pessimistic Time (b)`	The longest possible time an activity might take, considering potential obstacles, delays, or worst-case scenarios.	Accounts for potential risks and challenges, providing an upper bound for task duration in PERT.
`Project Crashing`	The process of reducing the overall project duration by accelerating specific activities, typically at an increased cost.	Essential for meeting aggressive deadlines or customer demands for earlier project completion.
`Critical Path`	The longest sequence of activities in a project network, determining the minimum project duration.	Identifying and managing the critical path is crucial because any delay on it directly delays the entire project.
`Hawthorne Effect`	The phenomenon where individuals improve their performance in response to being observed or given special attention.	Highlights the importance of managerial attention and interest in boosting team productivity and morale.
`Eligible Candidate`	An individual who possesses the formal qualifications and requirements for a job role.	Represents the theoretical fit based on credentials, but doesn't guarantee practical suitability.
`Suitable Candidate`	An individual who has the practical skills, aptitude, and temperament to effectively perform the job.	Represents the practical fit, emphasizing actual capability over formal qualifications for project success.
`Theory X`	A management style assuming employees are inherently lazy, lack ambition, and require close supervision, coercion, and control to perform.	Leads to a top-down, authoritarian management approach, potentially stifling initiative and motivation.

Term	Definition	Significance
`Theory Y`	A management style assuming employees are self-motivated, enjoy work, and seek responsibility, requiring encouragement and support.	Fosters a participative, empowering management approach, promoting creativity and commitment.

Detailed Analysis

1. PERT and Probabilistic Task Durations

While traditional Critical Path Method (CPM) assumes deterministic task durations, real-world development projects often involve significant uncertainty. PERT (Program Evaluation and Review Technique) addresses this by incorporating statistical estimates for activity durations. This approach acknowledges that exact durations are hard to predict, especially in novel or complex work.

- **Estimating Task Durations in PERT**: For each activity, three time estimates are provided:
- **Optimistic Time (a)**: The shortest possible time if everything goes perfectly.
- **Most Likely Time (m)**: The most probable time under normal circumstances.
- **Pessimistic Time (b)**: The longest possible time if significant obstacles or delays occur.
- **Calculating Estimated Time and Variance**: Based on these three estimates, the expected duration (Te) for an activity is calculated using a weighted average formula:
- `Te = (a + 4m + b) / 6`
- The variance (σ^2) of the activity duration is calculated as:
- `σ² = ((b - a) / 6)²`
- These statistical parameters allow for inferencing about project completion probabilities, similar to how deterministic durations are used in CPM. While the lecture provides an overview, it notes that specialized tools handle these calculations automatically.
- **Project Management Tools**: Several tools facilitate PERT and CPM calculations, aiding project managers in planning and tracking.

- **Gantt Project**: A simple, standalone desktop tool suitable for individual project managers, offering ease of use without extensive manuals. It automatically computes project characteristics and displays the critical path.
- **Redmine**: An open-source, server-based tool that supports collaboration, allowing multiple team members to view tasks, provide input, and track progress. This is beneficial for larger teams requiring shared access and real-time updates.
- The lecture encourages students to explore and become familiar with such tools, as proficiency in one often translates to easier adoption of others.

2. Project Crashing

Project crashing is a technique used to reduce the overall project duration, typically in response to customer demands for earlier completion. This process involves strategically accelerating certain tasks, which usually incurs additional costs.

- **Motivation for Crashing**: When a customer requests a project to be completed faster than the initially estimated duration (e.g., 5 months instead of 6), the project manager must find ways to shorten the schedule.
- **Crashing Methodology**:
- **Identify the Critical Path**: The first step is to determine the current critical path, as reducing tasks on this path directly shortens the project duration.
- **Identify Critical Tasks**: Within the critical path, identify all tasks that contribute to its length.
- **Determine Crashing Costs**: For each critical task, calculate the cost per day (or week) to expedite it and the maximum possible reduction in its duration. This involves deploying more resources (e.g., additional testers or coders).
- **Select the Cheapest Task**: Choose the critical task that offers the lowest cost per unit of time reduction.
- **Gradual Reduction**: Reduce the duration of the selected task incrementally (e.g., by one day or week).
- **Re-evaluate Critical Path**: After each reduction, re-evaluate the project network. The critical path may shift, or new critical paths may emerge.
- **Iterate**: Continue this process, focusing on the new critical path(s), until the target project duration is met or no further reduction is feasible or cost-effective.
- **Example of Crashing**: Consider a simple project with tasks A, B, C, D and a critical path A-B-C with a duration of 40 weeks.

- If task A can be reduced by 2 weeks at 500 rupees/week, task B by 3 weeks at 1800 rupees/week, task C by 2 weeks at 2000 rupees/week, and task D by 2 weeks at 2500 rupees/week.
- The project manager would initially choose task A because it is the cheapest to reduce (500 rupees/week).
- Reducing A by one week makes the critical path 39 weeks. Reducing it by another week makes it 38 weeks. The critical path remains A-B-C.
- Once A is fully crashed, the manager might then consider B, which is the next cheapest. However, as the critical path shortens, other paths might become critical, requiring simultaneous crashing of tasks on multiple critical paths to achieve further reduction.

3. Team Management and Organizational Behavior

Effective team management is crucial for project success, requiring daily effort from the project manager to track performance, motivate, provide feedback, and resolve conflicts. Understanding human behavior in the workplace is key to this.

- **The Hawthorne Effect**: Discovered in the 1920s at the Western Electric Hawthorne plant, experiments initially aimed to study the impact of lighting on productivity. Surprisingly, productivity increased regardless of whether light levels were raised or lowered. The conclusion was that the workers' awareness of being observed and receiving attention from management led to improved performance. This highlights that simply showing interest in a team can boost its productivity.
- **Selecting the Best People (Belbin's Distinction)**: When forming a team, it's vital to select individuals who are not just "eligible" but "suitable."
- **Eligible candidates** possess the required qualifications and credentials.
- **Suitable candidates** are those who can actually perform the job effectively.
- The danger lies in hiring eligible but unsuitable candidates, which harms both the project and the individual. The ideal, though challenging, scenario is to find suitable candidates who might not be formally eligible, as they may be more motivated and cost-effective.
- **Frederick Taylor's Scientific Management**: An early proponent of organizational behavior, Taylor recommended:
- Selecting the best people for the job.
- Instructing them in the best methods.
- Providing financial incentives based on work completed.

- While intuitive, applying "pieces of work completed" to software development can be complex due to its less tangible nature compared to manufacturing.
- **McGregor's Theory X and Theory Y**: These theories describe two contrasting managerial assumptions about employee motivation:
- **Theory X Manager**: Assumes employees are inherently lazy, dislike work, avoid responsibility, and need coercion, direction, and strict control to perform. Such managers often create environments where employees relax when supervision is absent.
- **Theory Y Manager**: Assumes employees are self-motivated, enjoy work (as natural as rest or play), seek responsibility, and are creative. These managers encourage and support their teams, fostering environments where work continues effectively even without direct supervision.
- **Characteristics of Software Developers**:
- **Proficiency Variation**: A 1968 study found a 1:25 ratio in the time taken by different programmers to code the same program, indicating a wide variance in individual proficiency. This underscores the importance of selecting highly competent developers.
- **Evolving Role**: While coding was a major activity in the past, modern software development involves many other tasks like design, testing, requirements analysis, and selecting reusable libraries. Therefore, a developer's "goodness" extends beyond just coding speed.
- **Social Aspects**: Early studies suggested software developers were less sociable, but later surveys found no significant social difference between IT workers and other professionals. This shift likely reflects the broader, more collaborative nature of contemporary software development.

Key Takeaways

- [ ] PERT is essential for project planning when task durations are uncertain, using optimistic, most likely, and pessimistic estimates.
- [ ] Project crashing is a strategic process to reduce project duration by accelerating critical tasks, always considering the associated costs.
- [ ] The critical path must be continuously re-evaluated during crashing, as it can shift or new critical paths may emerge.
- [ ] Managerial attention and interest (Hawthorne Effect) significantly boost team productivity and morale.
- [ ] Selecting "suitable" candidates who can perform the job is more crucial than merely "eligible" candidates with formal qualifications.

- [ ] Managers' assumptions about their team members (Theory X vs. Theory Y) profoundly influence leadership style and team dynamics.
- [ ] There is a wide variation in the proficiency of software developers, emphasizing the importance of careful selection and skill assessment.
- [ ] Modern software development involves a broad range of activities beyond just coding, requiring diverse skills and potentially more social interaction.

Conclusion

This lecture has provided a comprehensive overview of two critical dimensions of project management: adapting to scheduling uncertainties through PERT and strategically accelerating projects via crashing, alongside the indispensable art of team management. By understanding how to statistically model task durations, project managers can create more realistic schedules and proactively identify potential risks. The ability to effectively crash a project, by systematically identifying and accelerating critical tasks while managing costs, is a vital skill for meeting aggressive market demands and client expectations.

Equally important is the nuanced understanding of human behavior within project teams. Insights from the Hawthorne Effect, Belbin's candidate distinctions, Taylor's principles, and McGregor's theories offer a foundational framework for building, motivating, and leading high-performing teams. Recognizing that managerial attention, appropriate selection, and a positive view of employee motivation are powerful drivers of productivity allows project managers to cultivate an environment where individuals thrive and contribute optimally. Ultimately, successful project management hinges on a blend of robust technical methodologies and astute leadership, enabling projects to navigate complexities and achieve their objectives efficiently and effectively.